Phonological derivations are not harmonically improving: Evidence from Nazarene Arabic

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Abstract

Opacity poses a well-known challenge for the classical version of Optimality Theory (OT), which applies phonological processes in a fully parallel fashion. In response to the opacity challenge, a variety of extensions to the classical model have been proposed. Our focus is on extensions that incorporate a limited kind of serialism into OT: Harmonic Serialism and OT with Candidate Chains. The degree of serialism permitted by these theories is restricted by a property called *Harmonic Improvement*, according to which every derivational step creates a better output with respect to the single constraint ranking of the language. Harmonic Improvement rules out derivations of the form /A/ $\rightarrow$ |B| $\rightarrow$ [A] (also known as Duke-of-York derivations), because /A/ $\rightarrow$ |B| implies that B is better than A, while |B| $\rightarrow$ [A] implies the opposite. We present new data regarding the distribution of stress and vowel length in Nazarene Arabic, an understudied variety of Palestinian Arabic spoken in Nazareth. We argue that this distribution is best analyzed through a derivation of the form /A/ $\rightarrow$ |B| $\rightarrow$ [A], and thus poses a challenge to theories that obey Harmonic Improvement. Rule-based phonology and Stratal OT, which do not have this property, succeed in generating the pattern.

1 Introduction

In serial rule-based phonology (Chomsky and Halle, 1968), opacity – in the sense of Kiparsky 1971 – can naturally arise from the ordering of phonological rules. In Bedouin Hijazi Arabic, for example, a process that palatalizes an underlying /k/ before an /i/ typically gives rise to [kʲi] sequences on the surface. However, when palatalization interacts with a process that syncopates unstressed high vowels in open syllables, the trigger for palatalization can be removed from the surface, making it opaque (Al Mozainy, 1981). As can be seen in the serial derivation of [ħáːkʲmin] in (1), after palatalization has applied, syncope removes the conditioning environment for palatalization (there is no [i] after [kʲ] on the surface). The serial rule-based analysis attributes the application of palatalization without a trigger on surface to the ordering of palatalization before syncope.

(1) A rule-based ordering account of opacity in Bedouin Hijazi Arabic

UR	/ħaːkim/	/ħaːkim-in/
Palatalization	ħáːkʲim	ħáːkʲim-in
Syncope	-	ħáːkʲm-in
SR	[ħáːkʲim]	[ħáːkʲmin]
	'ruling-M.SG'	'ruling-M.PL'

In contrast to serial rule-based phonology, the classical version of Optimality Theory (OT; Prince and Smolensky 1993) is known to fail in generating opaque interactions, at least without additional mechanisms. This is due to a core property of OT that any alternation is triggered by a surface-oriented markedness constraint, which makes it difficult to trigger alternations with no support on the surface (McCarthy, 2007). As can be seen in the tableau in (2), a markedness constraint that motivates /i/ deletion in open syllables ($*iCV$) and a constraint that motivates /k/ palatalization ($*ki$) are both satisfied by /i/ deletion. Therefore, the correct candidate (2c), in which both processes have applied, is harmonically bounded by candidate (2d), in which only /i/ deletion has applied: candidate (2c) violates the faithfulness constraint that penalizes syncope (MAX) as well as the constraint that penalizes palatalization (IDENT(back)), whereas (2d) only violates the constraint that penalizes syncope, so no ranking of these constraints can work.

(2) The challenge of Parallel OT to account for opacity in BHA

	/ħaːkim-iːn/	$*iCV$	$*ki$	MAX	IDENT(back)
a.	ħaːkimiːn	*!	*		
b.	ħaːkʲimiːn	*!			*
c. ☹	ħaːkʲmiːn			*	*!
d. ☞	ħaːkmiːn			*	

Beyond this example of the challenge posed to OT by overapplication opacity, other kinds of opacity pose additional challenges to the classical OT model (Idsardi 2000; Vaux 2008; Baković 2011).

In response to the opacity challenge, a variety of extensions to the classical OT model have been proposed. Our focus in this paper is on extensions that incorporate a serial mechanism into OT in order to account for opacity: Stratal OT (Kiparsky, 2000, 2015; Bermúdez-Otero, 2011), Harmonic Serialism (HS; McCarthy 2000, 2008) and OT with Candidate Chains (OT-CC; McCarthy 2007). As we discuss in sections 3 and 4 in more detail, these theories differ in how radically they depart from classical OT: while all three apply phonological processes in a series of steps, they permit different degrees of serialism. Unlike Stratal OT, which assumes multiple constraint rankings, HS and OT-CC maintain the classical OT assumption of just one ranking and share a property called *Harmonic Improvement* (sometimes also called *Harmonic Ascent*; see Moreton 1999 and McCarthy 2000). According to Harmonic Improvement, every derivational step induces a change that creates an output that is "more harmonic" than the input of that step, i.e., every derivational step must generate a better form with respect to the single constraint ranking of the language.

Harmonic Improvement thus determines that if a language exhibits a change from /A/ → [B], this necessarily means that B is more harmonic than A. In such a language, a change from /B/ → [A] is subsequently ruled out because it contradictorily implies that A is more harmonic than B. Therefore, a significant prediction of Harmonic Improvement is that *Duke-of-York* derivations

(Pullum, 1976; McCarthy, 2003; Rubach, 2003; Odden, 2008) should not be attested.[1]

Duke-of-York derivations are derivations of the general form /A/ $\rightarrow$ |B| $\rightarrow$ [A], where an underlying /A/ is mapped to a different intermediate representation |B|, which is then mapped back to [A] on the surface. McCarthy (2003) showed that without a process that is crucially sandwiched between /A/ $\rightarrow$ |B| and |B| $\rightarrow$ [A], derivations that are analyzed as Duke-of-York derivations in rule-based phonology can also be accounted for in parallel OT (or course, without assuming that a process is undone). However, Duke-of-York derivations that involve an intermediate process can be generated by neither parallel OT or serial OT extensions that obey Harmonic Improvement.[2] McCarthy claimed that Duke-of-York derivations with an intermediate processes are unattested. To our knowledge, only a few attested patterns have been described since then: the interaction of multiple processes with voicing and devoicing in a variety of Japanese (Sasaki 2008), the interaction of tone with epenthesis and deletion in Arapaho (Gleim 2019), and the interaction of deletion with epenthesis and another deletion process in Modern Hebrew (Rasin 2022).

This paper presents new data regarding the distribution of stress and vowel length in Nazarene Arabic (NZA), an understudied variety of Palestinian Arabic spoken in Nazareth. We argue that this distribution is opaque and is best analyzed through a derivation of the form /A/ $\rightarrow$ |B| $\rightarrow$ [A], where the application of a vowel lengthening process affects stress assignment, only to have its outcome partly undone by a later process that shortens stressless vowels. Since stress assignment applies to the output of lengthening and since shortening applies to the output of stress, the ordering of stress between lengthening and shortening can be seen clearly.

We will argue that in order to generate the NZA pattern, a theory of phonology must have two architectural properties, stated in (3). First, the theory must allow for intermediate representations. Second, the theory must have enough freedom to allow phonological processes applying early in the derivation to be later undone.

(3) Architectural properties required to generate NZA
  a. Intermediate representations.
  b. Freedom to undo phonological processes.

We will make this argument by discussing the ability of several theories of phonology to generate the NZA pattern. We will show that only theories with both properties in (3) succeed on the pattern. As summarized in (4), rule-based phonology and Stratal OT satisfy both (3a) and (3b) and have no problem to generate the NZA pattern. Parallel OT, with or without extensions like transderivational faithfulness (Benua, 1997; Steriade, 1999), does not posit intermediate representations and thus violates both (3a) and (3b). HS and OT-CC allow intermediate representations but with restrictions that violate (3b). The latter three theories are unsuccessful on the NZA pattern. Our conclusion from this theoretical comparison will be that Harmonic Improvement is too restrictive, and that phonological derivations are not harmonically improving.

(4) A summary of the discussion in sections 2-4.

[1] More generally, Duke-of-York derivations involve a *fed-counterfeeding* interaction, which has been shown to pose a challenge to theories obeying Harmonic Improvement (Kavitskaya and Staroverov 2010; Staroverov 2014).

[2] McCarthy (2003) uses the terminology *vacuous* and *non-vacuous* Duke-of-York to refer to these two kinds of derivations. The pattern we will discuss in this paper will be shown to have similar theoretical implications to the non-vacuous kind.

	Property (3a)	Property (3b)	Successful on NZA
Rule-based phonology	✓	✓	✓
Parallel OT	✗	✗	✗
Harmonic Serialism	✓	✗	✗
OT-CC	✓	✗	✗
Stratal OT	✓	✓	✓

The paper is structured as follows. First, in section 2, we present the data and generalizations that will be used later in our theoretical discussion, as well as a rule-based analysis of the data. The rule-based analysis will make it easier to see the interaction between the relevant processes of NZA. Section 3 discusses the challenge raised by the NZA pattern for Parallel OT, HS and OT-CC. Section 4 presents a successful analysis within Stratal OT, which does not obey Harmonic Improvement. In section 5, we present further data from NZA, showing that the pattern discussed in this paper is fully general and holds without exception across the language.

2 Stress and vowel length in Nazarene Arabic

2.1 Basic generalizations

This section presents the core data on stress and vowel length in NZA, a variety of Palestinian Arabic spoken in Nazareth.[3] There has been much previous work on stress and vowel length in multiple colloquial dialects of Arabic, including varieties of Palestinian Arabic (Kenstowicz and Abdul-Karim, 1980; Abu-Salim, 1986; Younes, 1995; Watson, 2002; McCarthy, 2005, among others). The basic generalization regarding the distribution of stress is well established, and the basic facts about the distribution of vowel length have also been discussed.

The generalizations regarding stress and vowel length in NZA are not different from other varieties of Palestinian Arabic discussed in the literature. However, we chose to focus on one variety and specifically on NZA for two reasons. The first is that we wanted to verify that the relevant data discussed in previous literature on Palestinian Arabic are all generated by the same individual grammars (as opposed to data collected from different speakers or dialects with potentially inconsistent grammars; see de Lacy 2014 for discussion of this methodological point). The second reason is that we wanted to test the generality of the pattern beyond the examples discussed in the literature (below we present new data from dozens of morpheme combinations that have not been previously discussed). Since one of the authors is a native speaker of NZA, we were able to collect extensive data from this particular variety and confirm that they all come from the same grammar.

Turning to the generalizations, the stress pattern of NZA obeys the following algorithm (familiar from other dialects of Arabic; see Abdo, 1969; Brame, 1974; Watson, 2002):

(5) NZA stress pattern

[3]The name NZA has not been commonly used in the phonological literature, which typically makes a coarse distinction between two varieties of Palestinian Arabic – "rural" and "urban". One clear difference between so-called "rural" and "urban" dialects is the pronunciation of the Modern Standard Arabic voiceless uvular stop [q], often pronounced as the glottal stop [ʔ] in "urban" and as the velar stop [g] in "rural". The variety spoken in Nazareth shows a mix of [ʔ] and [q] (the choice depends on sociological factors) and so does not fit into the common coarse classification. We therefore refer to it distinctly as "NZA".

a. Stress the final syllable if superheavy (CVCC or CV:C).
b. Otherwise, stress the penultimate syllable if heavy (CVC or CV:) or word-initial.
c. Otherwise, stress the antepenultimate syllable.

The pattern is illustrated by the examples in (6). In (6a), the final syllable is a super-heavy CV:C syllable so it receives stress according to (5a). In (6b), the final syllable is a heavy CVC syllable, so (5a) does not hold. Instead, since the penultimate syllable is heavy, (5b) is met and penultimate stress is assigned. In (6c), the two final syllables are light CV syllables so neither (5a) nor (5b) are met and stress falls on the antepenultimate syllable by (5c). Beyond the examples in (6), the stress pattern is fully general and exceptionless.[4]

(6) a. mak.tab.téːn 'two libraries'
b. mak.táb.tek 'your.f.sg library'
c. mák.ta.be 'library'

Vowel length can be unpredictable in stressed syllables, as illustrated by pairs of vocalic templates that differ only with respect to the length of one of the vowels:

(7) a. CáCaC e.g., katab 'he wrote'
b. Cá:CaC e.g., kaːtab 'he corresponded'

In unstressed syllables, however, the distribution of vowel length is always predictable. Several analyses have been proposed to account for this distribution in terms of a process of unstressed-vowel shortening. Abu-Salim (1986) proposed that unstressed vowels in open syllables are shortened before stressed vowels (e.g, [báːb] '*door*' → [babéːn] '*two doors*'), but not before unstressed vowels (e.g, [ʕáːlam] '*world*' → [ʕaːlaméːn] '*two worlds*'). He proposed a metrical analysis that enforces the shortening of pretonic vowels, and also correctly accounts for an opaque interaction between shortening and syncope (e.g, /ʃaːtir-iːn/ → [ʃaːtríːn] '*smart*.pl'). In a later study, Younes (1995) discussed northern rural dialects of Palestinian Arabic that are not completely compatible with Abu-Salim's (1986) generalizations. Younes suggested that pretonic shortening applies to vowels in open syllables only when they precede long stressed vowels (but not when they precede short stressed vowels). The NZA facts seems to be compatible with his analysis.[5]

Consider the verbal paradigms of [ʃáːf] '*he saw*' in NZA in (8). The table in (8a) shows the basic past inflection of the verb with all possible subject markers. The paradigms in (8b) show how three of the subjects markers can be combined with the object clitics of the language. The 3.sg.m column shows that the stem's vowel remains long before an object clitic when that vowel is stressed (e.g., [ʃáːfni] '*he saw me*'). The 3.sg.f column shows that the stem's vowel also remains long before a stressed CVC syllable (e.g., [ʃaːfátni] '*she saw me*', as in Younes 1995, but not as in Abu-Salim 1986). The 3.pl column shows that the stem's vowel is shortened in open syllables before long stressed vowels (e.g., [ʃafúːni] '*they saw me*').

[4]Though not always surface true, as stress can become obscured on the surface through vowel epenthesis. There has been extensive research on the interaction of stress and epenthesis in Arabic in the context of theory comparison (see, e.g., Broselow 1982, 2000; Kiparsky 2000). However, this interaction is independent of the theoretical discussion of the present paper, so we leave it aside.

[5]See Abu-Salim (1986) and Younes (1995) for further observations about vowel shortening in other environments that will not play a role in our paper.

(8) Verbal paradigms of [ʃáːf] '*he saw*'[6,7]

a. Past subject inflections

Subject		
Infl	Affix	
1.sg	-t	ʃúfit
1.pl	-na	ʃúfna
2.sg.f	-ti	ʃúfti
2.sg.m	-t	ʃúfit
2.pl	-tu	ʃúftu
3.sg.f	-at	ʃáːfat
3.sg.m	∅	ʃáːf
3.pl	-u	ʃáːfu

b. Past subject markers with object clitics

Object		Subject inflection		
Infl	Affix	3.sg.m	3.sg.f	3.pl
1.sg	-ni	ʃáːfni	ʃaːfátni	ʃafúːni
1.pl	-na	ʃáːfna	ʃaːfátna	ʃafúːna
2.sg.f	-ek/-ki	ʃáːfek	ʃáːfatek	ʃafúːki
2.sg.m	ak/-k	ʃáːfak	ʃáːfatak	ʃafúːk
2.pl	-ku	ʃáːfku	ʃaːfátku	ʃafúːku
3.sg.f	-ha	ʃáːfha	ʃaːfátha	ʃafúːha
3.sg.m	-o/-h	ʃáːfo	ʃáːfato	ʃafúː
3.pl	-hen	ʃáːfhen	ʃaːfáthen	ʃafúːhen

Subject and object markers may be followed by the negation morpheme /-ʃ/. Adding negation to the verbs in table (8b) reveals an environment where shortening overapplies to vowels that do not immediately precede a stressed syllable with another long vowel, as shown in (9). For example, while shortening applied right before a stressed long vowel in [ʃafúːni] 'they saw me' in (8b), the same vowel is shortened in [ma ʃafuníːʃ] '*they did not see me*', even though the following syllable [fu] has a short, unstressed vowel. In section 2.2 we will analyze this overapplication as a cyclic effect, where shortening applies in an intermediate representation (ʃaːfúːni) before the negation morpheme is added.

(9) Past inflections of [ʃáːf] with object clitics and negation

Object		Subject inflection		
Infl	Affix	3.sg.m	3.sg.f	3.pl
1.sg	-ni	ma ʃaːfníːʃ	ma ʃaːfatníːʃ	ma ʃafuníːʃ
1.pl	-na	ma ʃaːfnáːʃ	ma ʃaːfatnáːʃ	ma ʃafunáːʃ

Additional data show that the distribution of vowel length is also sensitive to the word boundary and to internal morpheme boundaries. Consider the verbal paradigms of [nísi] '*he forgot*' in (10). The paradigm in (10a) consists of the past subject inflections of the verb, and the paradigm in (10b) consists of all the combinations of the 1.pl subject affix with object clitics. The verb [nísi] is conventionally analyzed with the root $\sqrt{n.s.j}$ whose glide consonant is only realized before vowels. Otherwise, the stem ends with a vowel.

[6]"Infl = Inflection. Our use of "Affix" does not make the traditional distinction between affixes and clitics.

[7]The 3.sg.m object marker [-h] can be deleted, as in /ʃaːf-u-h/ → [ʃafúː] 'they saw him'.

(10) Verbal paradigms of [nísi] '*he forgot*'

a. Past subject inflections:

Subject		Root
Infl	Affix	n.s.j
1.sg	-t	nsíːt
1.pl	-na	nsíːna
2.sg.f	-ti	nsíːti
2.sg.m	-t	nsíːt
2.pl	-tu	nsíːtu
3.sg.f	-at	nísjat
3.sg.m	∅	nísi
3.pl	-u	nísju

b. Past object clitics for 1.pl subjects:

Object		Root
Infl	Affix	n.s.j
1.sg	-ni	nsináːni
1.pl	-na	nsináːna
2.sg.f	-ki	nsináːki
2.sg.m	-k	nsináːk
2.pl	-ku	nsináːku
3.sg.f	-ha	nsináːha
3.sg.m	-h/-o	nsináː
3.pl	-hen	nsináːhen

Across the data in (10), stressless word-final vowels are short, illustrating the following generalization about NZA:

(11) Stressless word-final vowels are always short.

The second generalization concerns the length of vowels that immediately precede a (word-internal) morpheme boundary. As can be seen in (10b), some pre-boundary vowels are stressed and long (e.g., the penultimate vowel in [nsi-n**áː**-ni]), while others are stressless and short (e.g., the antepenultimate vowel in [ns**i**-náː-ni]). Our generalization about these data is only an approximation and can be stated as in (12). The reason that we can only provide an approximation at this descriptive stage is that, as we will argue later in the paper, the analysis of vowel length requires reference to an intermediate representation, making the distribution of vowel length difficult to state as a surface generalization. Nevertheless, we find an approximated generalization to be helpful before seeing an analysis.

(12) Before a morpheme boundary, a vowel is long in a position where a long vowel would receive stress, and is otherwise short.

To clarify this generalization, consider the word [nsíː-na] '*we forgot*' in (10a). The stem-final vowel [i] immediately precedes a morpheme boundary. A long vowel in this position should receive stress according to clause (5b) in the stress algorithm. Therefore, this vowel is long (and is also stressed). In [nsi-náː-ha] '*we forgot her*' in (10b), the pre-boundary vowel [a] is in a position where a long vowel would be stressed, and is therefore long. The preceding pre-boundary vowel [i] should not receive stress in this position, and is short.

Importantly, hypothetical words that do not obey the generalization in (12) are ungrammatical, as shown in (13). A descriptively adequate phonological grammar must rule out these systematic gaps.

(13) Examples of systematic gaps in NZA

a. Before a morpheme boundary, a vowel cannot surface as short in a position where a long vowel would receive stress:

- *nsí-na-ku 'we forgot you.pl'
- *nsí-na-ku-ʃ 'we didn't forget you.pl'

b. Before a morpheme boundary, unstressed vowels cannot surface as long:

- *nsiː-náː-ku 'we forgot you.pl'
- *nsi-naː-kúː-ʃ 'we didn't forget you.pl'

The generalization in (12) can also be seen clearly in the imperfect paradigm (e.g, [btínsu] '*you.pl are forgetting*', [btinsúːha] '*you.pl are forgetting her*', and [btinsuháːʃ] '*you.pl are not forgetting her*'). In fact, the generalization holds without exception across dozens of different morpheme combinations, which to our knowledge are all possible NZA configurations in which the generality of the pattern can be tested. This is shown in section 5.

2.2 Analysis within rule-based phonology

In this section we present a rule-based analysis for the NZA pattern described above. We will provide an NZA grammar that can generate all of the data, as well as rule out the ungrammatical systematic gaps. We will show that such a grammar requires both intermediate representations and the freedom to undo phonological processes. Consider first the simple grammar in (14), which has two rules: STRESS, which assigns stress according to the algorithm in (5), and SHORTENING, which shortens vowels in open syllables before stressed long vowels.

(14) Grammar:

1. STRESS: assign stress according to (5).[8]
2. SHORTENING: V→ [-long] / ___ CV́ː

Consider the derivation of [btinsuháːʃ] '*you.pl are not forgetting her*'. As can be seen in the left column in (15), as long as all morpheme-final vowels are underlyingly long, such a grammar can generate the correct surface form. First, since the final syllable is super-heavy, stress is assigned to it. Only then, vowel shortening applies to the pretonic vowel and the correct output is derived. In contrast, the right column in (15) shows that if the morpheme-final vowels are underlyingly short, stress is assigned to the wrong vowel, shortening does not apply, and an ungrammatical output is derived.

(15) A rule-based analysis of NZA without LENGTHENING

UR	/btins-uː-haː-ʃ/	/btins-u-ha-ʃ/
STRESS	btins-uː-háː-ʃ	btíns-u-ha-ʃ
SHORTENING	btins-u-háː-ʃ	-
SR	[btinsuháːʃ]	*[btínsuhaʃ]

As McCarthy (2005) already pointed out, and exemplified in (15), shortening by itself fails to rule out systematic gaps in Arabic, as we must take into account multiple underlying representations (URs). McCarthy's OT solution for this problem will be discussed later. Within rule-based phonology, a simple way to rule out this kind of systematic gaps is to add an early lengthening rule, given in (16), which lengthens any vowel followed by a morpheme boundary.

(16) LENGTHENING: V → [+long] / ___+

[8]A formal stress assignment rule can be stated as follows: V → [+stress] /___$C_0((VC)VC_0^1)\#$ (cf. Brame 1974).

The derivations in (17) show that the three rules correctly derive [btins-u-háː-ʃ] regardless of the underlying length specification of morpheme-final vowels (which alternate between long and short).[9]

(17) Derivation of [btins-u-háː-ʃ] from URs with different length specifications

UR	/btins-u-ha-ʃ/	/ btins-u-haː-ʃ/	/ btins-uː-ha-ʃ/	/ btins-uː-haː-ʃ/
LENGTHENING	btins-uː-haː-ʃ	btins-uː-haː-ʃ	btins-uː-haː-ʃ	btins-uː-haː-ʃ
STRESS	btins-uː-háː-ʃ	btins-uː-háː-ʃ	btins-uː-háː-ʃ	btins-uː-háː-ʃ
SHORTENING	btins-u-háː-ʃ	btins-u-háː-ʃ	btins-u-háː-ʃ	btins-u-háː-ʃ
SR	[btins-u-háː-ʃ]	[btins-u-háː-ʃ]	[btins-u-háː-ʃ]	[btins-u-háː-ʃ]

There are two further things to account for: the length of word-final vowels, and the overapplication of shortening in some environments. These will not play a role in our argument, but are discussed here to present a more complete analysis. First, as shortening applies only to pretonic vowels, the grammar does not rule out words with a stressless long final vowel (e.g., *[btinsúːhaː]), which contradict the generalization in (11). Such surface forms can be ruled out straightforwardly by adding a rule that shortens word-final stressless vowels:

(18) FINAL SHORTENING: $\underset{\text{[-stress]}}{\text{V}} \rightarrow \text{[-long]} / \_\_\#$

The second thing to account for is the opaque shortening in forms such as [ma ʃafuníːʃ] *'they did not see me'*. In such configurations, the long vowel in /ʃaːf/ is shortened even though there is no conditioning environment for shortening on the surface – the vowel does not immediately precede a long stressed vowel. This can be accounted for by applying the rules cyclically, as also proposed by Abu-Salim (1986) and Watson (2002). Put simply, a cyclic application of the rules means that they apply after each stage of morphological suffixation. In the derivation of [ma ʃafuníːʃ], for example, the stem's vowel /aː/ is shortened before a stressed long vowel in the intermediate representation |ʃafúːni|. Only then, the negation marker is added, stress is reassigned to the ultimate syllable and removes the environment of shortening from the surface. This is illustrated in (19).[10]

(19) A cyclic application of the rules in the derivation of [ma ʃafuníːʃ]

/ma ʃaːf-u-ni-ʃ/ $\xrightarrow[\text{1}^{\text{st}}\text{ cycle}]{}$ |ʃáːfu| $\xrightarrow[\text{2}^{\text{nd}}\text{ cycle}]{}$ |ʃafúːni| $\xrightarrow[\text{3}^{\text{rd}}\text{ cycle}]{}$ [ma ʃafuníːʃ]

After establishing the rule-based analysis of the pattern, consider again the leftmost column in (17), which we have isolated in (20).

(20) Derivation of [btins-u-háː-ʃ] from a UR with morpheme-final short vowels

UR	/btins-u-ha-ʃ/
LENGTHENING	btins-uː-haː-ʃ
STRESS	btins-uː-háː-ʃ
SHORTENING	btins-u-háː-ʃ
SR	[btins-u-háː-ʃ]

[9] As will be discussed briefly in section 5, there is some evidence suggesting that the UR of the stem is in fact /btinsaj/, and that the underlyingly final vowel and glide are deleted. Whether these deletion processes take place or not, they do not interact with the pattern that we are interested in, so we leave them aside.

[10] This derivation assumes that internal morpheme boundaries are deleted at the end of every cycle, as can be enforced by Bracket Erasure (see Pesetsky 1979 and Kiparsky 1982a).

In this derivation, STRESS is crucially sandwiched between SHORTENING and LENGTHENING, while SHORTENING partially removes the effects of LENGTHENING (/u/ $\rightarrow$ |uː| $\rightarrow$ [u]). This derivation manifests the architectural properties that we will argue are require to derive the NZA pattern, given in (3) and repeated in (21).

(21) Architectural properties required to generate NZA
   a. Intermediate representations.
   b. Freedom to undo phonological processes.

Indeed, as we will see in the next sections, the NZA data pose a challenge to phonological architectures that do not posit these properties.

3 The challenge for 'Harmonic Improvement'

3.1 The challenge for Parallel OT

According to Harmonic Improvement, a single constraint ranking determines the entire phonological mapping and every derivational step must generate a better output with respect to the constraint ranking. Parallel OT obeys Harmonic Improvement with one derivational step. In what follows, we will first introduce and discuss the challenge that the NZA pattern poses for Parallel OT. Then, we will get to the challenge for serial varieties of OT that obey Harmonic Improvement (HS and OT-CC).

We have seen that according to the rule-based analysis, stress in NZA is sensitive to vowel length in an intermediate representation, after vowel lengthening and before vowel shortening. Parallel OT computes stress and vowel length in parallel, without intermediate representations (as in, for example, McCarthy's 2005 analysis, to be reexamined in section 3.1.2), and thus has a problem assigning stress to the right position. In particular, without intermediate representations, the challenge for Parallel OT is to choose the output in (22a) over the incorrect but prosodically-well-formed alternative in (22b) without causing trouble elsewhere. To show the problem, we will consider various choices for stress constraints and their ranking.

(22) Correct output vs. problematic alternative
   a. [btins-u-háː-ʃ] (correct)
   b. *[btins-úː-ha-ʃ] (incorrect)

3.1.1 Standard stress constraints

Consider first a standard choice of stress constraints from Prince (1990) and Prince and Smolensky (1993) and a ranking of these constraints, given in (23), that could generate stress as in (5) after certain additions and refinements that will be left aside to keep the discussion simple.

(23) $\text{WSP}_{\text{Super-heavy}} \gg$ NONFINALITY $\gg \text{WSP}_{\text{Heavy}} \gg$ EXTNONFINALITY $\gg$ ...

The constraints in (23) are EXTNONFINALITY, which penalizes stress on either of the final two syllables, $\text{WSP}_{\text{Heavy}}$, which requires heavy syllables (CVC, CV:) to be stressed, NONFINALITY, which penalizes stress on the final syllable, and $\text{WSP}_{\text{Super-heavy}}$, which requires super heavy syllables

(CVCC, CV:C) to be stressed. The ranking (again, after some additions and refinements) would ensure that stress on the ante-penultimate syllable is the default, but this default can be overridden if the penultimate syllable is heavy (in which case stress will be penultimate) or if the final syllable is superheavy (in which case stress will be final).

Because NONFINALITY is violated by final stress, this ranking favors non-final stress over final stress, which means that among the two output candidates in (22), it favors the incorrect candidate, as schematized in (24). Here we use URs with morpheme-final long vowels to illustrate the problem (of course, according to the principle of Richness of the Base (Prince and Smolensky, 1993; Smolensky, 1996), the correct output needs to be selected given any underlying specification of length. We will return to this point later on).

(24)

		/btinsuː-haː-ʃ/	$\text{WSP}_{\text{Super-heavy}}$	NONFINALITY	...
a.	☞	btins-úː-ha-ʃ			...
b.	☹	btins-u-háː-ʃ		*!	...

For a Parallel OT analysis to work, some other constraint (or constraints) $\mathbb{C}$ that favors the correct candidate (24b) over candidate (24a) must outrank NONFINALITY. With such a constraint in place, candidate (24b) will win, as illustrated schematically in the following tableau:

(25)

		/btinsuː-haː-ʃ/	$\mathbb{C}$	$\text{WSP}_{\text{Super-heavy}}$	NONFINALITY	...
a.		btins-úː-ha-ʃ	$\cdots$ ***!** $\cdots$			...
b.	☞	btins-u-háː-ʃ	$\cdots$		*	...

Which constraint can $\mathbb{C}$ be? We will go over several options that do not work before mentioning an option that works technically and discussing its limitations. First, $\mathbb{C}$ cannot be simple markedness constraints that are independently motivated by the distribution of length in NZA. Consider the constraints in (26):

(26)
- $*\breve{V}+$ - assign * for every short vowel that is immediately followed by a morpheme boundary.
- $*V{:}C\acute{V}{:}$ - assign * for every long vowel in open syllable that precedes a stressed long vowel.

The first constraint in (26) can motivate the lengthening of vowels before morpheme boundaries, and the second can motivate pretonic shortening. These independently motivated constraints, however, cannot account for the choice of the locus of stress and lengthening. As can be seen in (27), the candidates from (24) are equally harmonic with respect to these markedness constraints.

(27)

	/btinsuː-haː-ʃ/	$*V{:}C\acute{V}{:}$	$*\breve{V}+$	...
a.	btins-úː-ha-ʃ		*	...
b.	btins-u-háː-ʃ		*	...

$\mathbb{C}$ cannot be a familiar input-output faithfulness constraint either. As mentioned above, the correct pattern should be generated given URs with any specification of vowel length. Therefore, any attempt to favor [btins-u-háː-ʃ] over *[btins-úː-ha-ʃ] that refers to the underlying length in /btins-uː-haː-ʃ/ will not work for a different UR where the same vowels are short. For the same reason, faithfulness to underlying stress will be of no help.

Consider next transderivational faithfulness constraints based on Output-to-Output Correspondence as in Benua 1997, Steriade 1999, and much related work. Transderivational faithfulness constraints can require similarity between the output of a derived form and its base, where the base can be approximately defined as the free-standing output form that contains the largest subset of the grammatical features of the derived form (Kager 1999). For example, the constraint OO-DEP-μ prohibits a mora in the derived form that is not also present in the base. Such constraints do not seem to help in selecting candidate [btins-u-háː-ʃ], because the incorrect candidate *[btins-úː-ha-ʃ] would be more similar to the base ([btins-úː-ha]). The challenge for Parallel OT is to be able to choose an output candidate that is less similar to the base, so the logic of transderivational faithfulness would lead it astray.[11]

Consider now the possibility that $\mathbb{C}$ is a more complex markedness constraint that is not independently motivated in NZA. To understand the degree of specificity required from this constraint, consider the words in (28), which have an underlying long vowel.

(28)	a.	miʃwáːr-ak	'your.m.sg walk'	(cf. miʃwar-éːn 'two walks')
	b.	ʔəstaʃáːr-ek	'he consulted you.f.sg'	(cf. ʔəstaʃar-úː-ni 'they consulted me')

Given the prosodic well-formedness of these words in NZA, $\mathbb{C}$ would have to penalize candidate *[btins-úː-ha-ʃ] without also penalizing (28a)-(28b). This is because, the choice of candidate the [btins-u-háː-ʃ] over *[btins-úː-ha-ʃ] under a ranking with such a high-ranking markedness constraint $\mathbb{C}$ would teach us that $\mathbb{C}$ can be satisfied by stressing the final syllable and at the same time lengthening a final underlyingly-short vowel. If $\mathbb{C}$ also penalized (28a)-(28b), the fact that $\mathbb{C}$ can be repaired by stressing the final syllable and lengthening the final vowel would mean that the ungrammatical *[miʃwar-áːk] should win over [miʃwáːr-ak]), and that the ungrammatical *[ʔəstaʃar-éːk] should win over [ʔəstaʃáːr-ek]), contrary to fact. To distinguish *[btins-úː-ha-ʃ] from (28a)-(28b), $\mathbb{C}$ could be a markedness constraint that refers to morpheme boundaries, which are the main surface difference between *[btins-úː-ha-ʃ] and (28a)-(28b). A markedness constraint that penalizes *[btins-úː-ha-ʃ] but not [btins-u-háː-ʃ] or (28a)-(28b) and refers to morpheme boundaries is the one in (29):

(29) $^{*}\breve{V} + \underbrace{\{C, V\}^{*}}_{No\ +} \#$: assign * for a short vowel at the end of the penultimate morpheme of the word (whatever the morphology of the word is)

With this constraint in place, the correct candidate is the winner:

(30)			/btinsuː-haː-ʃ/	$^{*}\breve{V} + \{C, V\}^{*}\#$	$\text{WSP}_{\text{Super-heavy}}$	NonFinality	...
	a.		btins-úː-ha-ʃ	*!			...
	b.	☞	btins-u-háː-ʃ			*	...

The major problem with (29) is not its complexity or its lack of independent motivation in NZA, but rather that it makes pathological typological predictions that theories without it can avoid. Examples of processes that (29) can generate (by appropriately ranking it with respect to faithfulness constraints) include the following:

[11] Existing variants of transderivational faithfulness (e.g., Albright 2002; Stanton and Steriade 2014) that can pressure a derived form to be more similar to other bases (including non-local bases) do not seem to help either, because as more suffixes are added, there is a vowel goes back and forth between being short and long ([btíns-**u**], [btins-**úː**-ha], [btins-**u**-háː-ʃ]). If [btins-u-háː-ʃ] has a short [u] because of similarity to the base [btíns-u], the challenge would be to explain why [btins-úː-ha] would not also have a short [u] for the same reason and surface as the ungrammatical *[btíns-u-ha] instead.

(31) Pathological typological predictions made by (29)

a. Delete the vowel in the penultimate morpheme of the word if that vowel is short.

b. Insert a consonant after the vowel in the penultimate morpheme of the word if that vowel is short.

Such processes violate a broader generalization regarding attested phonological processes that has been implicitly assumed to hold by phonologists for decades. By including both + and # in its description, (29) causes a direct violation of the following phonological universal:

(32) **Universal** (entailed by the proposals of Chomsky and Halle 1968, Pesetsky 1979, Kiparsky 1982b, Orgun and Inkelas 2002, and Bermúdez-Otero 2011):
A phonological process cannot refer to both a word boundary and an internal morpheme boundary.[12]

If the universal in (32) is correct, constraints like (29) should be excluded from the space of possible constraints in OT.[13]

We can conclude that an analysis of the NZA pattern in terms of (29) results in unwelcome typological predictions in different degrees of severity. The pathologies in (31) already seem like bad enough predictions, and until counterexamples to the universal in (32) are found, the problem for (29) is even more general. The rule-based analysis from section 2.2 as well as the Stratal OT analysis that will be presented in section 4 do not suffer from the same problem. By positing intermediate representations, those analyses account for the distribution of stress in a simple way and do not have to resort to rules or constraints like (29).

3.1.2 A reexamination of McCarthy's (2005) analysis

McCarthy (2005) proposes a Parallel OT analysis for the distribution of stem-final vowels in Cairene Arabic, which is similar to the NZA pattern discussed here. We will show that his analysis succeeds on pre-boundary vowels in words that have one suffix, but breaks down when there is more than one suffix.

McCarthy points out that shortening by itself is not sufficient to rule out systematic gaps in Cairene Arabic, akin to the NZA gaps in (13). Therefore, he proposes that an underlying long vowel is shortened word-finally, but also that that an underlying short vowel is deleted in the same position. His analysis uses the constraints in (33), as well as the stress constraints NONFINALITY and WSP that were presented above.

(33) McCarthy's (2005a) constraints

- MAX-μ - assign * for every mora in the input that has no correspondent in the output.
- MAX-V: - assign * for every long vowel in the input that has no correspondent in the output.

[12] Processes that refer to both a word boundary and an internal morpheme boundary were ruled out as early as SPE by the Bracket Erasure convention. While some versions of Bracket Erasure were under debate in the 1980s, a weaker version was defended by Orgun and Inkelas (2002), from which (32) still follows.

[13] Note that the problem with (29) is independent of whether OT constraints are innate or acquired. As long as (29) can be stated in the grammar and be ranked above the relevant faithfulness constraints, the patterns in (31) would be generated.

- FIN-C - assign * for every word final vowel.
- OO-DEP-V - assign * for every vowel in the output that has no correspondent in the base's output.

To see how the analysis works, consider first the unsuffixed forms in (34) and (35).

(34) McCarthy's (2005a) derivation of a stem-final short vowel

		/takaba/	NONFIN	WSP	MAX-Vː	OO-DEP-V	FIN-C	MAX-μ
a.	☞	tákab						*
b.		tákaba					*!	

(35) McCarthy's (2005a) derivation of a stem-final long vowel

		/takabaː/	NONFIN	WSP	MAX-Vː	OO-DEP-V	FIN-C	MAX-μ
a.		tákab			*!			**
b.	☞	tákaba					*	*
c.		takabáː	*!					
d.		tákabaː		*!				

Tableau (34) shows that an underlyingly-short word-final vowel is deleted, in order to satisfy FIN-C, the markedness constraint that requires words to end with a consonant. Tableau (35) shows that an underlyingly-long vowel in the same position is shortened. If the long vowel surfaced as long, it would have caused a violation of the stress markedness constraints NONFINALITY and WSP. But because MAX-Vː outranks FIN-C, the optimal solution is to shorten the vowel rather than deleting it.

Consider now the same vowel-final stem before a suffix. McCarthy's analysis ensures that some grammatical surface form is generated no matter what the underlying length of the stem-final vowel is. This is shown in tableaux (36) and (37).

(36) McCarthy's (2005a) derivation of a suffixed stem with a short final vowel

		/takaba-haː/	NONFIN	WSP	MAX-Vː	OO-DEP-V	FIN-C	MAX-μ
	base:	[tákab]						
a.	☞	takábha					*	*
b.		takábaha				*!	*	

(37) McCarthy's (2005a) derivation of a suffixed stem with a long final vowel

		/takabaː-haː/	NONFIN	WSP	MAX-Vː	OO-DEP-V	FIN-C	MAX-μ
	base:	[tákaba]						
a.	☞	takabáː-ha					*	*
b.		takaba-háː	*!				*	*
c.		takabáː-haː		*!			*	

Tableau (36) shows the derivation where the stem-final vowel is underlyingly short. In this case, OO-DEP-V kicks in and rules out candidate (36b) because the base [tákab] (the output of (34)) does not end with a vowel. On the other hand, if the stem-final vowel is underlyingly long, as in tableau (37), OO-DEP-V is irrelevant because the base [tákaba] (the output of (35)) ends with a vowel. The stem-final vowel remains long and receives stress on the surface.

McCarthy (2005) suggested that his analysis should also work for longer morpheme configurations, but this is not always the case, as can be seen by considering the NZA example [btins-u-háː-ʃ] we used so far. We show this in tableau (38), which uses McCarthy's constraints and the UR /btins-uː-haː-ʃ/. The incorrect candidate (38a) is favored by McCarthy's constraints as it has strictly fewer violations.

(38) McCarthy's (2005a) analysis fails on words with two pre-boundary vowels

		/btins-uː-haː-ʃ/	NonFin	WSP	Max-Vː	OO-Dep-V	Fin-C	Max-μ
	base:	[btinsúːha]						
a.	☞	btins-úː-ha-ʃ						*
b.	☹	btins-u-háː-ʃ	*!					*

To summarize this section, McCarthy (2005) has shown that his analysis succeeds on derivations that consist of a stem and one suffix. However, since his analysis is a Parallel OT analysis, is still suffers from the problem we discussed earlier: without intermediate representations, it cannot choose the correct position of the stressed long vowel in words with longer morpheme configurations like [btins-u-háː-ʃ]. In what follows we will show that this problem also holds if we consider different stress constraints.

3.1.3 Alternative stress constraints

The discussion so far has assumed the stress constraints in (23) that make non-final stress the default. A default of non-final stress is not an accident of that set of constraints, but rather a general consequence of how complementarity is typically analyzed within OT (see Baković 2013 for much relevant discussion): the occurrence of an element in a specific environment (e.g., voiced obstruents only intervocalically) is enforced using a markedness constraint penalizing the absence of that element in that environment (e.g., *VTV in the case of intervocalic voicing). The reverse – penalizing the element in the elsewhere set of environments – would often require a complex list of constraints with potentially problematic typological consequences (e.g., constraints penalizing voiced obstruents word-initially in the case of intervocalic voicing). The stress pattern of NZA is no different: the effect of weight on stress (final stress only on super-heavy syllables, otherwise penultimate stress only on heavy syllables, otherwise antepenultimate stress) motivates a constraint hierarchy that makes antepenultimate stress (the elsewhere case) the default, and that uses constraints like $\text{WSP}_{\text{Heavy}}$ and $\text{WSP}_{\text{Super-heavy}}$ to push stress rightwards precisely when heavier syllables follow.

Suppose, however, that some reasonable constraints could be devised that account for the complementarity in NZA stress by making rightmost stress the default and by deflecting stress leftwards away from syllables that are too light.[14] We will now show that the NZA pattern poses a problem for Parallel OT even under such constraints. By doing that, we will have shown that the problem for Parallel OT discussed in the previous sections is more general and did not arise because of the choice of the stress constraints in (23).

[14]One conceivable set of ranked constraints that can generate the NZA stress pattern and has this property is the following: $\text{NonFinality}_{\text{Light}}$ (penalize a final light stressed syllable), $\text{NonFinality}_{\text{Heavy}}$ (penalize a final heavy stressed syllable), $\text{ExtNonFinality}_{\text{Light}}$ (penalize a penultimate light stressed syllable), and AlignR (stress is aligned to the right), where AlignR is dominated by the three other constraints.

Consider again the tableau for /btins-uː-haː-ʃ/ from (24), repeated in (39) with two arbitrary stress-markedness constraints, which this time make final stress the default. Since final stress is the default, the incorrect candidate (39a) receives a fatal violation and the correct candidate (39b) is now chosen.

(39)

		/btins-uː-haː-ʃ/	Stress-M_1	Stress-M_2	...
a.		btins-úː-ha-ʃ		*!	...
b.	☞	btins-u-háː-ʃ			...

This analysis, while correctly generating [btins-u-háː-ʃ], creates a new problem elsewhere. Given Richness of the Base, the correct candidate (39b) must be generated even when the underlying form has a long stem-final vowel followed by a short suffix-final vowel, i.e., /btins-uː-ha-ʃ/. For candidate (39b) to win over candidate (39a) as an output of /btins-uː-ha-ʃ/, the pressure for final stress must be strong enough to enforce final stress even at the expense of lengthening and shortening underlying vowels. This, in turn, makes the incorrect prediction that words like the ones mentioned above in (28), such as [miʃwáːr-ak] (from underlying /miʃwaːr-ak/) and [ʔəstaʃáːr-ɛk] (from underlying /staʃaːr-ɛk/), which have the same prosodic profile as candidate (39a), should surface with final stress and a final long vowel, contrary to fact:

(40)

a.

		/btins-uː-ha-ʃ/	Stress-M_1	Stress-M_2	IDENT(long)	...
a.		btins-úː-ha-ʃ		*!		...
b.	☞	btins-u-háː-ʃ			**	...

b.

		/miʃwaːr-ak/	Stress-M_1	Stress-M_2	IDENT(long)	...
a.	☹	miʃwáːr-ak		*!		...
b.	☞	miʃwar-áːk			**	...

To save Parallel OT, then, the challenge is to find a grammar that generates the correct [btins-u-háː-ʃ] without also over-generating ungrammatical alternatives to words like [miʃwáːr-ak].

The conclusion for Parallel OT is that the theory is unable to account for the NZA pattern. The failure is due to the full parallelism of the theory and its rejection of intermediate representations. In what follows we will see that adding serialism to OT is insufficient to solve the problem, as long as processes cannot be undone in the course of the derivation.

3.2 The challenge for Harmonic Serialism and OT-CC

HS (McCarthy 2000, 2008) is a serial version of OT. Similarly to Parallel OT, an HS grammar includes one set of ranked, violable constraints. Differently from Parallel OT, computation is serial. GEN, which generates output candidates, is limited to changing the input by one atomic change at a time (epenthesis, deletion, feature change, etc.). At each step of the serial derivation, EVAL selects the most harmonic candidate as the output, which then serves as the input for the next step. GEN and EVAL then loop until convergence.

Because of its serial nature, HS has intermediate representations that it can use to correctly account for a subset of the NZA pattern: the interaction of lengthening and stress. However, assuming the standard architectural assumptions of the theory, HS obeys Harmonic Improvement because every step of the derivation must improve the form with respect to a single constraint ranking. As we will see, given Harmonic Improvement, it is not possible to undo earlier changes in the derivation, as needed in NZA.

Let us first see how intermediate representations allow HS to correctly account for the interaction of stress and lengthening. By ranking the markedness constraint that triggers lengthening ($^*\breve{V}+$) above whatever constraints enforce the correct distribution of stress (from now on they will be merged into the cover constraint STRESS), lengthening will correctly apply first. To see how this works, consider the first step of the derivation of [btins-úː-ha] from a UR with short vowels, /btins-u-ha/, given in (41).[15]

(41)

Step I		/btins-u-ha/	$^*\breve{V}+$	STRESS	$^*V{:}C\acute{V}{:}$	ID(stress)	ID(long)
a.		btins-u-ha	*!	*			
b.	☞	btins-uː-ha		*			*
c.		btíns-u-ha	*!			*	

Here, there are three candidates that differ from the input by at most once change. Candidate (41a) is the fully-faithful candidate. In candidate (41b) lengthening has applied, and in candidate (41c) stress has been assigned. Candidates (41a) and (41c) both violate the highest-ranked $^*\breve{V}+$, so they lose to candidate (41b), whose violation is of the lower-ranking constraint STRESS.

The output of the first step serves as the input to the second step in (42). Now that $^*\breve{V}+$ has been resolved by lengthening, the derivation can attend to the lower-ranking STRESS constraint and resolve it by correctly assigning stress in candidate (42b), [btins-úː-ha], which will be the input to the third step. In the third step (not shown), all markedness constraints that outrank faithfulness constraints have no remaining violations, so [btins-úː-ha] will win again and the derivation will converge on it as the final output.

(42)

Step II		/btins-uː-ha/	$^*\breve{V}+$	STRESS	$^*V{:}C\acute{V}{:}$	ID(stress)	ID(long)
a.		btins-uː-ha		*			
b.	☞	btins-úː-ha				*	

In HS, the ranking of the constraints determines the order of operations. The ranking $^*\breve{V}+ \gg$ STRESS ensured that lengthening will apply before stress assignment, because candidates that do not violate the higher-ranking $^*\breve{V}+$ receive priority earlier in the derivation. The same reasoning means that in order for stress to apply before shortening in NZA, STRESS must outrank the markedness constraint that triggers shortening, $^*V{:}C\acute{V}{:}$. By transitivity, $^*\breve{V}+$ must outrank $^*V{:}C\acute{V}{:}$.

It is easy to see that the ranking $^*\breve{V}+ \gg {^*V{:}C\acute{V}{:}}$ causes a problem for the HS account of NZA. According to this ranking, a long vowel before a morpheme boundary is less marked than a short vowel in the same position. This ranking thus incorrectly blocks the shortening of stressless vowels before a morpheme boundary, such as the vowel [u] of [btins-u-háː-ʃ]. In the HS derivation of the input /btins-u-ha-ʃ/, and assuming the ranking $^*\breve{V}+ \gg$ STRESS $\gg {^*V{:}C\acute{V}{:}}$, lengthening and stress will first apply in the right order, yielding the intermediate representation |btins-uː-háː-ʃ| after the third step of the derivation (i.e., after two applications of lengthening and one application of stress). In the fourth step, shown in (43), shortening of the stressless /uː/ will be incorrectly blocked by $^*\breve{V}+$, and the derivation will converge on the ungrammatical *[btins-uː-háː-ʃ] instead of the correct [btins-u-háː-ʃ].

[15]For space reasons, we shorten the constraint label IDENT to ID in this tableau and in the tableaux below.

(43)

Step IV		/btins-uː-háː-ʃ/	$^*\breve{V}+$	STRESS	$^*V{:}C\acute{V}{:}$	ID(stress)	ID(long)
a.	☞	btins-uː-háː-ʃ			*		
b.	☹	btins-u-háː-ʃ	*!				

As in the case of Parallel OT, simple revisions of the constraint set in (43) do not seem to help: similarly to Parallel OT, HS can adopt non-local complex constraints along the lines of (29), but these would create the same typological problems they did for Parallel OT.

Overall, the challenge for HS comes from the requirement that a single constraint ranking is optimized serially: if it had been possible to change the ranking of the constraints in the course of the derivation, the problem could have been solved by promoting $^*V{:}C\acute{V}{:}$ after lengthening and stress had been assigned. This problem is not specific to HS. Other serial theories assuming Harmonic Improvement, such as OT-CC (McCarthy, 2007), to which we now turn, fail on the NZA pattern for the same reason.

In OT-CC, a candidate is not just a single surface form, but rather a chain of forms that represents a serial derivation. The first member of the chain is the UR and every following member (except the last one) is an intermediate representation that differs from the previous member by a single change, just as in HS. The last member of the chain represents the surface form. The chain must obey Harmonic Improvement, meaning that every non-initial step of the chain must be better than the previous step with respect to the single constraint ranking of the language. An important component of OT-CC is PREC constraints (from 'precedence'), which can penalize candidate chains with an undesirable order of faithfulness violations. The introduction of PREC constraints into the theory is what provides OT-CC with additional generative power compared to HS, and allows it to generate instances of opacity that HS fails on. According to McCarthy, PREC constraints only come into play after the complete set of chains has been determined (McCarthy 2007, p. 99), so we can discuss the candidate chains first.

A consequence of the Harmonic Improvement requirement on candidate chains is that the Duke-of-York derivation /u/ → |uː| → [u] in NZA is an illegal chain. This is because, as we have seen in the discussion of HS, a single constraint ranking cannot lengthen and then shorten the same vowel. An illegal chain for the example [btinsuháːʃ] is shown in (44).

(44) A Duke-of-York chain for [btinsuháːʃ] (illegal)
<btinsu-ha-ʃ, btinsuː-ha-ʃ, btinsuː-haː-ʃ, btinsuː-háː-ʃ, btinsu-háː-ʃ>

Since Duke-of-York chains are ruled out, any chain that leads to the correct output should not lengthen the vowel /u/ at all. The challenge for OT-CC is therefore to favor candidate (45a) over alternative candidates like (45b) and (45c). In the tableau in (45), below every chain, we show (in curly brackets) the ordered set of faithfulness-constraint violations, along with the position in the string where the violation is incurred (e.g., "IDENT(long)@8" means that there has been a violation of IDENT(long) in the 8th segment of the string). The tableau illustrates that OT-CC, just like HS, favors the incorrect candidate (45c).

(45) OT-CC fails to account for NZA[16]

[16] For space reasons, the candidate chains do not include the faithful first step, <btins-u-ha-ʃ>.

	/btins-u-ha-ʃ/	$^*\breve{V}+$	STRESS	$^*V{:}C\acute{V}{:}$	ID(stress)	ID(long)
a. ☹	btinsuháːʃ <btins-u-haː-ʃ, btins-u-háː-ʃ> {IDENT(long)@8, IDENT(stress)@8}	*!			*	*
b.	btinsúːhaʃ <btins-uː-ha-ʃ, btins-úː-ha-ʃ> {IDENT(long)@6, IDENT(stress)@6}	*!			*	*
c. ☞	btinsuːháːʃ <btins-uː-ha-ʃ, btins-uː-haː-ʃ, btins-uː-háː-ʃ> {IDENT(long)@6, IDENT(long)@8 IDENT(stress)@8}			*	*	**

Regular markedness and faithfulness constraints will be of no help, for reasons we have already discussed. The addition of PREC constraints, which is OT-CC's innovation, cannot help either. PREC constraints are constraints of the form PREC(A,B), where A and B are faithfulness constraints. PREC(A,B) demands that every violation of B in the chain is preceded and is not followed by a violation of A. In (45), candidates (45a) and (45b) share the very same sequence of faithfulness violations, so there is no PREC constraint that can distinguish between them.

We can conclude that the intermediate representations of HS and the chains of OT-CC are not sufficient to generate the NZA pattern because of Harmonic Improvement. Other serial OT theories that assume a single one constraint ranking, such as Serial Markedness Reduction (Jarosz 2014), are similarly expected to fail on the NZA pattern as long as they obey Harmonic Improvement and do not posit the freedom to undo phonological processes. This point will be key to the next section, where we will show that unlike HS and OT-CC, Stratal OT succeeds on the NZA pattern.

4 Stratal OT analysis

This section presents a schematic analysis of the basic NZA data presented in section 2 within Stratal OT (Kiparsky 2000, 2015, Bermúdez-Otero 2011). There are multiple versions of Stratal OT that differ from each other with respect to properties such as the number of strata, which strata are cyclic, and so on. But the properties that enable Stratal OT to generate the NZA pattern are shared by all versions of the theory we know of: not restricted by Harmonic Improvement, Stratal OT posits at least two strata with different constraint rankings, with the following properties:

(46) Properties of Stratal OT

a. The output of constraint evaluation in one stratum serves as the input to constraint evaluation in a following stratum.

b. A ranking of $C_1 \gg C_2$ between two constraints in one stratum may be replaced with the opposite ranking $C_2 \gg C_1$ in another stratum.

The combination of these two properties allows Stratal OT to easily avoid the problem that [btinsuháːʃ] posed for Harmonic Improvement. Namely, that the lengthened [uː] in *[btinsuːháːʃ] could not have been shortened after applying lengthening and stress assignment. Under Stratal OT, the possibility of reversing the constraint ranking between strata means that a Stratal OT analysis can locate lengthening and stress assignment in one stratum and shortening in a separate, later stratum. More concretely, consider the grammar in (47), which uses the same constraints from section 3.

(47) Stratal OT grammar for NZA (approximation)

a. Stratum I: $^*\breve{V}+$, STRESS $\gg$ IDENT(stress), IDENT(long) $\gg$ $^*V{:}C\acute{V}{:}$

b. Stratum II: $^*V{:}C\acute{V}{:}$, STRESS $\gg$ IDENT(stress), IDENT(long) $\gg$ $^*\breve{V}+$

In the ranking of Stratum I, the markedness constraints that enforce lenghtening and stress, $^*\breve{V}+$ and STRESS, outrank the corresponding faithfulness constraints. In addition, $^*\breve{V}+$ outranks $^*V{:}C\acute{V}{:}$. Therefore, in this stratum, short vowels will be lengthened before a morpheme boundary and stress will be assigned based on a representation in which all pre-boundary vowels are long. This is shown in the tableau in (48), where the optimal candidate is [btins-uː-háː-ʃ].[17]

(48) Stratum I: lengthening and stress are active.

		/btins-u-ha-ʃ/	$^*\breve{V}+$	STRESS	IDENT(stress)	IDENT(long)	$^*V{:}C\acute{V}{:}$
a.		btíns-u-ha-ʃ	*!*				
b.		btins-úː-ha-ʃ	*!			*	
c.		btins-u-háː-ʃ	*!			*	
d.	☞	btins-uː-háː-ʃ				**	*

The output of Stratum I, [btins-uː-háː-ʃ], serves as the input to Stratum II. In Stratum II, as shown in (49), the markedness constraint that triggers shortening, $^*V{:}C\acute{V}{:}$, has been promoted above both $^*\breve{V}+$ and IDENT(long), so the long /uː/ in the input is correctly shortened. On this analysis, IDENT(stress) favors the correct candidate (49b) over candidate (49c), in which stress shift has occurred. The outcome is that the correct output [btinsuháːʃ] is chosen.

(49) Stratum II: shortening is active.

		/btins-uː-háː-ʃ/	$^*V{:}C\acute{V}{:}$	STRESS	IDENT(stress)	IDENT(long)	$^*\breve{V}+$
a.		btins-uː-háː-ʃ	*!				
b.	☞	btins-u-háː-ʃ				*	*
c.		btins-úː-ha-ʃ			*!*	*	*

As mentioned above, there are multiple different versions of Stratal OT, which differ in their assumptions about the number of strata, the cyclic status of each stratum, how exactly the phonology and morphology are interleaved, etc. Those difference are not significant for the main argument of this paper, because all versions can change the constraint ranking in the course of the derivation and avoid the problem that NZA poses for Harmonic Improvement.

Nevertheless, choosing between these versions and attributing strata I and II in the above analysis to concrete strata is an interesting question. We believe that it should be answered not just based on the NZA puzzle we focused on here, but also based on a variety of facts about Palestinian Arabic that have figured prominently in the literature, such as the well-known opaque interaction of stress and syncope (Brame, 1974; Kenstowicz and Abdul-Karim, 1980; Kiparsky, 2000), the interaction of stress and epenthesis (Broselow 1982, 2000; Kiparsky 2000), and the cyclic effect observed in section 2.[18] We therefore leave a full analysis of NZA within Stratal OT for future research.

[17] Given the high ranking of $^*\breve{V}+$, the winning candidate in this tableau will be the optimal candidate regardless of the underlying length specification of the morpheme-final vowels.

[18] For example, as we noted in footnote 10, some version of Bracket Erasure might be necessary, even in Stratal OT.

5 The pattern is fully general

Our goal in this section is to show that the interaction of stress and vowel length in NZA as described in Subsection 2.1 is fully general and goes beyond the examples presented before. Consider the word [btins-u-háː-ʃ] '*you are not forgetting her*', which exemplifies the generalization in (12) as a vowel preceding a morpheme boundary is long only where it would receive stress and otherwise short. The relevant property of this word is that it includes a sequence of three morphemes M_1, M_2, M_3 such that M_1 and M_2 are vowel-final and the entire sequence occurs word-finally. The existence of M_3 and the fact that the sequence occurs word-finally causes stress to fall on the final vowel of M_2. The occurrence of the vowel-final M_1 in the sequence results in a configuration where some vowel occurs before a morpheme boundary in a position where it does not receive stress. In what follows, we describe all the different morpheme sequences in NZA verbs we know of that have this property.

The general structure of the NZA verb is given in (50). The portion that is relevant for our purposes is the portion that includes the stem and the suffixes. Suffixes can include subject agreement markers, object clitics, and the negation suffix, in this order.

(50) Structure of the NZA verb
(Prefixes)-STEM-(SUBJECT.AGR)-(OBJECT.CLITIC)-(NEG)

Morphemes that can be vowel-final are the stem, the perfective subject agreement markers -ti '2.sg.f', -na '1.pl', -tu '2.pl', and -u '3.pl', the imperfective subject agreement markers -i '2.sg.f' and -u '2.pl' or '3.pl', and the direct object clitics -ni '1.sg', -ki '2.sg.f', -ka '2.sg.m', -ha '3.sg.f', -na '1.pl', and -ku '2.pl'. Together we get four types of morpheme combinations with the relevant properties, which are listed in the following table. In this table and below we keep using the stem of the verb 'forget' for illustration, and the negation suffix is -ʃ.

(51) Morpheme sequences with two adjacent vowel-final morphemes preceded by a word-final suffix.

Type	M_1 (vowel-final)	M_2 (vowel-final)	M_3	Example	Alternative
I	SUBJECT.AGR	OBJECT.CLITIC	NEG	btins-u-háː-ʃ	*btins-úː-ha-ʃ
II	STEM	OBJECT.CLITIC	NEG	btinsa-níː-ʃ	*btins-áː-ni-ʃ
III	STEM	SUBJECT.AGR	OBJECT.CLITIC	nsi-túː-ha	*ns-íː-tu-ha
IV	STEM	SUBJECT.AGR	NEG	nsi-náː-ʃ	*ns-íː-na-ʃ

The full range of examples is given below – a total of 74 different morpheme combinations – divided by type.

In Type I examples, which we present in two parts below for reasons of formatting, the stem is followed by a vowel-final subject agreement marker, then a vowel-final object clitic, then the negation marker (the stem is also vowel final in some of these examples, but here we focus on the suffixes and the relationship between them and disregard the stem). The examples below are all the possible combinations of this type (with the verb 'forget'). The running example in the paper, [btins-u-háː-ʃ] '*you are not forgetting her*', is of this type, so each of these examples share the attributes that will have theoretical implications in section 3.

Here we should note that additional processes might be involved in the derivation of the surface stem of verbs like '*forget*', depending on what the UR of the stem is, a question that there is no agreement on in the literature on Arabic. Vowel-final stems seem to come from tri-consonantal roots with a final glide (e.g., the root of 'forget' is n.s.j) which surfaces in some environments (e.g., [nís.ju] '*they forgot*') but not in others (as in most examples in this paper). If the UR of the stem includes a final glide, then at least a process that deletes or alters the final glide would have to take place in addition to lengthening before the object clitic (e.g., /binsaj-ni/ → [binsáːni] '*he is forgetting me*'). As far as we can tell, the possibility of an underlying glide and a process that deletes or alters it does not affect the main argument of the paper.

(52) Type I examples – part 1 (total: 18/36 examples in part 1)

a. ma nsi-ti-níː-ʃ 'You.f didn't forget me'
b. ma nsi-ti-kíː-ʃ 'You.f didn't forget you.f'
c. ma nsi-ti-káː-ʃ 'You.f didn't forget you.m'
d. ma nsi-ti-háː-ʃ 'You.f didn't forget her'
e. ma nsi-ti-náː-ʃ 'You.f didn't forget us'
f. ma nsi-ti-kúː-ʃ 'You.f didn't forget you.pl'
g. ma nsi-na-níː-ʃ 'We didn't forget me'
h. ma nsi-na-kíː-ʃ 'We didn't forget you.f'
i. ma nsi-na-káː-ʃ 'We didn't forget you.m'
j. ma nsi-na-háː-ʃ 'We didn't forget her'
k. ma nsi-na-náː-ʃ 'We didn't forget us'
l. ma nsi-na-kúː-ʃ 'We didn't forget you.pl'
m. ma nsi-tu-níː-ʃ 'You.pl didn't forget me'
n. ma nsi-tu-kíː-ʃ 'You.pl didn't forget you.f'
o. ma nsi-tu-káː-ʃ 'You.pl didn't forget you.m'
p. ma nsi-tu-háː-ʃ 'You.pl didn't forget her'
q. ma nsi-tu-náː-ʃ 'You.pl didn't forget us'
r. ma nsi-tu-kúː-ʃ 'You.pl didn't forget you.pl'

(53) Type I examples: part 2 (total: 18/36 examples in part 2)

a. ma nisj-u-níː-ʃ 'They.pl didn't forget me'
b. ma nisj-u-kíː-ʃ 'They.pl didn't forget you.f'
c. ma nisj-u-káː-ʃ 'They.pl didn't forget you.m'
d. ma nisj-u-háː-ʃ 'They.pl didn't forget you.m'
e. ma nisj-u-náː-ʃ 'They.pl didn't forget us'
f. ma nisj-u-kúː-ʃ 'They.pl didn't forget you.pl'
g. btins-i-níː-ʃ 'You.f are not forgetting me'
h. btins-i-kíː-ʃ 'You.f are not forgetting you.f'
i. btins-i-káː-ʃ 'You.f are not forgetting you.m'

j. btins-i-háː-ʃ 'You.f are not forgetting her'
k. btins-i-náː-ʃ 'You.f are not forgetting us'
l. btins-i-kúː-ʃ 'You.f are not forgetting you.pl'
m. btins-u-níː-ʃ 'You.pl are not forgetting me'
n. btins-u-kíː-ʃ 'You.pl are not forgetting you.f'
o. btins-u-káː-ʃ 'You.pl are not forgetting you.m'
p. btins-u-háː-ʃ 'You.pl are not forgetting her'
q. btins-u-náː-ʃ 'You.pl are not forgetting us'
r. btins-u-kúː-ʃ 'You.pl are not forgetting you.pl'

In Type II examples, the sequence of relevant morphemes consists of a vowel-final stem, followed by a vowel-final object clitic, which is in turn followed by the negation marker. Vowel lengthening applies to stem-final vowels before an object clitic (e.g., [nisíː-ni] 'he forgot me') so these examples show the same pattern as before: the only long vowel is the rightmost vowel that precedes a morpheme boundary (in this case, the vowel of the object clitic).

(54) Type II examples (total: 12 examples)
a. ma nisi-níː-ʃ 'he didn't forget me'
b. ma nisi-kíː-ʃ 'he didn't forget you.f'
c. ma nisi-káː-ʃ 'he didn't forget you.m'
d. ma nisi-háː-ʃ 'he didn't forget her'
e. ma nisi-náː-ʃ 'he didn't forget us'
f. ma nisi-kúː-ʃ 'he didn't forget you.pl'
g. binsa-níː-ʃ 'he is not forgetting me'
h. binsa-kíː-ʃ 'he is not forgetting you.f'
i. binsa-káː-ʃ 'he is not forgetting you.m'
j. binsa-háː-ʃ 'he is not forgetting her'
k. binsa-náː-ʃ 'he is not forgetting us'
l. binsa-kúː-ʃ 'he is not forgetting you.pl'

In Type III examples, the sequence of relevant morphemes consists of a vowel-final stem, followed by a vowel-final subject agreement marker, which is in turn followed by an object clitic. Type III has the same configuration that was shown in (10b) but the examples below show that the pattern is exceptionless for all vowel-final subject markers.

(55) Type III examples (total: 23 examples)
a. nsi-tíː-ni 'you.f forgot me'
b. nsi-tíː-ki 'you.f forgot you.f'
c. nsi-tíː-ha 'you.f forgot her'
d. nsi-tíː-(h) 'you.f forgot him'
e. nsi-tíː-na 'you.f forgot us'

f. nsi-tíː-ku 'you.f forgot you.pl'
g. nsi-tíː-hen 'you.f forgot them.pl'
h. nsi-náː-ni 'we forgot me'
i. nsi-náː-ki 'we forgot you.f'
j. nsi-náː-k 'we forgot you.m'
k. nsi-náː-ha 'we forgot her'
l. nsi-náː-(h) 'we forgot him'
m. nsi-náː-na 'we forgot us'
n. nsi-náː-ku 'we forgot you.pl'
o. nsi-náː-hen 'we forgot them'
p. nsi-túː-ni 'you.pl forgot me'
q. nsi-túː-ki 'you.pl forgot you.f'
r. nsi-túː-k 'you.pl forgot you.m'
s. nsi-túː-ha 'you.pl forgot her'
t. nsi-túː-(h) 'you.pl forgot him'
u. nsi-túː-na 'you.pl forgot us'
v. nsi-túː-ku 'you.pl forgot you.pl'
w. nsi-túː-hen 'you.pl forgot them'

Finally, in Type IV examples, a vowel-final stem is followed by a subject marker and then negation.

(56) Type IV examples (total: 3 examples)
a. ma nsi-tíː-ʃ 'you.f didn't forget'
b. ma nsi-náː-ʃ 'we didn't forget'
c. ma nsi-túː-ʃ 'you.pl didn't forget'

In sum, the generalization regarding stress and vowel length in (12) holds as expected in all relevant morpheme combinations that we could think of, without exception. This suggests that the pattern is fully general and is not limited to particular morphemes or morpheme combinations.

6 Conclusion

This paper discussed a pattern from an understudied variety of Palestinian Arabic spoken in Nazareth. We have argued that phonological theory must adopt two architectural properties in order to successfully generate this pattern: intermediate representations and the freedom to undo processes in the course of the derivation. Rule-based phonology and Stratal OT, which have these two properties, were successful, but Parallel OT, HS and OT-CC, which do not have both, did not succeed. A summary of our theoretical discussion, which was shown in (4), is repeated here in (57).

(57) Summary of the discussion in Sections 2-4.

	Property (3a)	Property (3b)	Successful on NZA
Rule-based phonology	✓	✓	✓
Parallel OT	✗	✗	✗
Harmonic Serialism	✓	✗	✗
OT-CC	✓	✗	✗
Stratal OT	✓	✓	✓

The NZA pattern joins other instances of Duke-of-York derivations discussed in the literature and mentioned in the introduction, such as those from Japanese (Sasaki, 2008), Arapaho (Gleim, 2019), and Modern Hebrew (Rasin, 2022). The increasing number of such patterns suggests that NZA is not an isolated example, and that Duke-of-York derivations are a real property of natural language that should be accounted for by theories of phonology.

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